# User Guide

## 1.1 Introduction

This section presents the user guide for the NutriVision Food Nutrition Estimator web application. The purpose of this guide is to help end users operate the deployed system without requiring any technical knowledge. The application enables users to upload an image of an Indian meal, use a ₹10 coin as a scale reference, and obtain an estimated nutritional profile of the food items present in the image.

The system performs end-to-end analysis by combining food segmentation, food classification, geometric portion estimation, weight estimation, and nutrition lookup. In addition, the application provides an interpretable reasoning panel that explains how the estimated weight was derived. This makes the system both practical for real-world use and transparent in its predictions.

## 1.2 Accessing the Application

The NutriVision application is deployed as a browser-based web interface and can be accessed using the following URL:

**Application Link:**
**https://7860-01knjbx5j5gnvmttvxykcmx4bh.cloudspaces.litng.ai/**

The application does not require user authentication or account registration. It can be opened directly in any modern web browser such as Google Chrome, Microsoft Edge, or Mozilla Firefox.

### Minimum Usage Requirements

To use the system effectively, the user should have the following:

- A stable internet connection
- A clear food image in JPG, JPEG, or PNG format
- A visible ₹10 coin placed near the food item for scale calibration
- A modern browser for smooth rendering of the interface

## 1.3 Application Interface Overview

When the application is opened, the user is presented with a simple and intuitive interface titled Food Nutrition Estimator. The main interface contains the following components:

- Application Title: *Food Nutrition Estimator*

- Usage Instruction Banner: prompts the user to upload a food image with a ₹10 coin as a size reference
- Pipeline Overview:
  *SAM3 segmentation → ConvNeXtV2 classification → PCA geometry → LLM weight estimation → Nutrition lookup*
- Upload Panel: allows the user to upload a food image
- Analyse Button: initiates the complete prediction pipeline
- Detected Food Items Panel: displays the segmented and labeled food item(s)
- Pipeline Status Checklist: shows the completion of major processing stages
- Nutrition Summary Table: displays detailed nutritional values for the detected food item(s)
- LLM Weight Reasoning Section: provides interpretability for the estimated weight and geometry assumptions

The interface is designed to guide the user from image upload to final nutritional output in a single workflow.

**Figure 1.1** shows the main landing interface of the application.

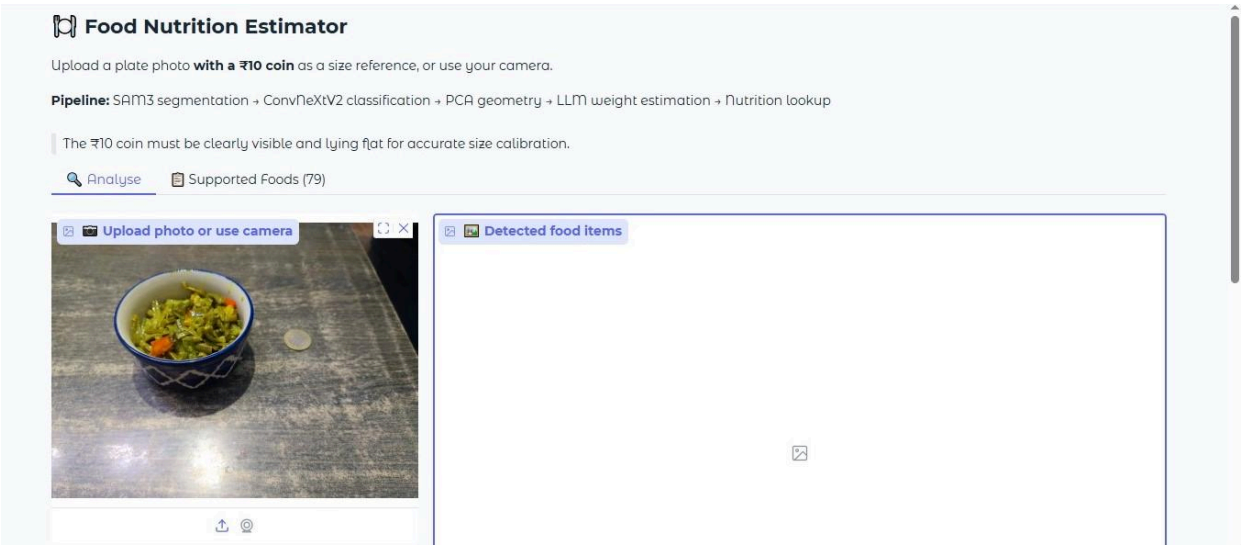



**Figure 1.1: NutriVision application landing screen showing the upload panel, pipeline description, and user instructions.**

# 1.4 Step-by-Step Usage Instructions

## 1.4.1 Step 1: Open the Web Application

The user should begin by opening the deployed NutriVision web application in a browser using the provided link:

https://7860-01knjbx5j5gnvmttvxykcmx4bh.cloudspaces.litng.ai/

After the page loads, the user will see the main interface with the upload panel and the analysis controls.

### 1.4.2 Step 2: Prepare the Food Image

Before uploading an image, the user should ensure that the food is clearly visible and that the image is suitable for accurate analysis. The following input conditions are recommended:

- The food should be clearly visible and not heavily occluded
- The image should be sharp and not blurry
- The image should be taken under good lighting conditions
- The photo should preferably be captured from a top-down or near top-down angle
- The background should be simple and not visually cluttered

The application accepts the following image formats:

- JPG
- JPEG
- PNG

The user may either upload an existing image from the device or capture a fresh image using the built-in camera option provided by the application.

### 1.4.3 Step 3: Place a ₹10 Coin for Scale Calibration

A key requirement of the NutriVision system is the use of a ₹10 coin as a real-world size reference. Before capturing or uploading the image, the user must place a ₹10 coin next to the food item in the same frame.

This step is essential because the system uses the coin to convert pixel-based measurements into real-world dimensions. Since the ₹10 coin has a known diameter of approximately 2.7 cm, it allows the system to estimate:

- The physical dimensions of the food item
- The approximate volume of the food
- The estimated weight in grams
- The final nutritional values based on portion size

### Guidelines for Correct Coin Placement

For best results, the ₹10 coin should satisfy the following conditions:

- It should be fully visible in the image
- It should be placed close to the food item

- It should be lying flat on the same plane as the food
- It should not be blurred, tilted excessively, or partially cropped

If the coin is absent or not clearly visible, the system may still identify the food, but the portion estimation and nutrition values may be less reliable.

### 1.4.4 Step 4: Upload the Image

The user should click the “Upload photo” option in the left-side panel of the application.

The image can then be provided in one of two ways:

1. Upload from Device: Select an existing image file from the local device
2. Use Camera: Capture a new image directly using the browser camera interface

After the image is uploaded, the application displays a preview in the upload panel. The user should verify that both the food item and the ₹10 coin are clearly visible before proceeding.

### 1.4.5 Step 5: Start the Analysis

Once the image has been uploaded successfully, the user should click the “Analyse” button.

Clicking this button initiates the complete NutriVision inference pipeline. The application automatically performs the following sequence of operations:

1. SAM3 Segmentation – separates the food region from the background
2. Coin Scale Detection – identifies the ₹10 coin for size calibration
3. Food Classification – predicts the food category using the trained classifier
4. PCA Geometry Estimation – estimates geometric dimensions from the segmented mask
5. Weight and Nutrition Estimation – calculates approximate weight and nutrient values
6. LLM Weight Reasoning – explains the assumptions used in the final estimate

No manual intervention is required during this process.

## 1.5 Processing Pipeline Feedback

After the analysis begins, the application provides visual feedback by displaying a checklist of completed processing stages. This improves usability by allowing the user to confirm that each component of the pipeline has executed successfully.

The application shows stage completion indicators such as:

- SAM3 Segmentation
- Coin Scale Detection
- Food Classification

- PCA Geometry
- Weight & Nutrition Estimation

This pipeline transparency is useful for both demonstration and practical usage, as it clearly communicates the internal stages of the system to the user.

## 1.6 Viewing the Detected Food Output

After processing is completed, the right-side Detected food items panel displays the analyzed image with the segmented food region highlighted.

This panel typically shows:

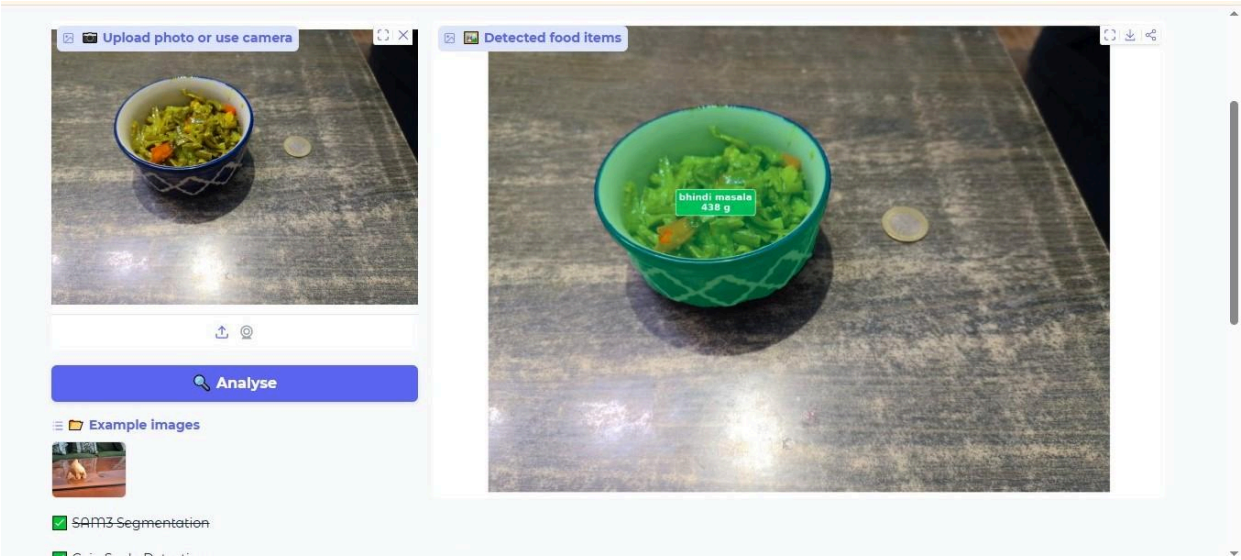
- The food item region overlaid with a segmentation mask
- The predicted food label
- The estimated weight displayed directly on the image

For example, in the sample application output, the system correctly identifies the uploaded food as bhindi masala and overlays the prediction on the segmented region.

This visual output helps the user verify that the system has:

- detected the correct region
- assigned the correct food label
- estimated the portion in a meaningful way

**Figure 1.2** shows the detection and segmentation result produced by the application.




## 1.7 Understanding the Results

Once the pipeline is complete, the application generates the final results in three main forms:

1. Detection Summary
2. Nutrition Summary Table
3. LLM Weight Reasoning Section

Each of these outputs is described below.

### 1.7.1 Detection Summary

The application first displays a short summary indicating the number of detected food items and the approximate total calories.

A typical summary appears in the following format:

- 1 food item detected
- 485 kcal total

This serves as a quick overview before the detailed nutritional breakdown is presented.

### 1.7.2 Nutrition Summary Table

The most important output of the application is the Nutrition Summary table. This table provides the nutritional profile of each detected food item based on the estimated weight and the nutrition lookup database.

The table contains the following fields:

- Food
- Weight (g)
- Calories (kcal)
- Carbohydrates (g)
- Protein (g)
- Fat (g)
- Free Sugar (g)
- Fibre (g)
- Sodium (mg)
- Calcium (mg)
- Iron (mg)

This table enables the user to understand both the portion size and the nutritional composition of the meal.

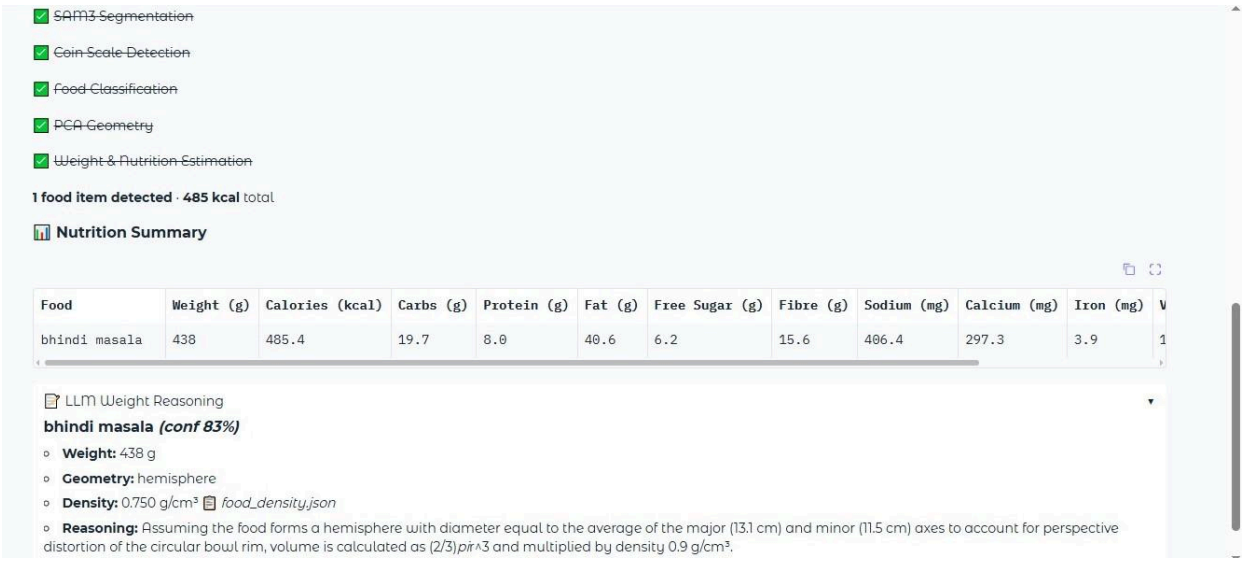
### 1.7.3 LLM Weight Reasoning Section

Below the nutrition table, the application provides an **LLM Weight Reasoning** panel. This section is important because it explains how the system arrived at the estimated weight, making the model's output more transparent and interpretable.

The panel typically displays:

- **Predicted food name**
- **Confidence score**
- **Estimated weight**
- **Assumed geometric shape**
- **Density value used**
- **Reasoning behind the estimate**

**Figure 1.3** should present the final nutritional output and reasoning interface.




# 1.8 Best Practices for Accurate Results

To maximize the quality of predictions and nutritional estimates, users should follow the recommendations below:

- Capture the image in good lighting to reduce shadows and visual noise
- Use a top-down or near top-down camera angle whenever possible
- Ensure that the ₹10 coin is fully visible and lying flat
- Place the coin close to the food item for better scale calibration
- Avoid blurry or low-resolution images
- Ensure the food item occupies a meaningful portion of the frame
- Avoid strong perspective distortion or extreme side angles
- If multiple items are present, try to keep them visually separated
- Avoid heavy overlap between different dishes

- Prefer a clean and simple background to improve segmentation quality

Following these best practices improves the reliability of segmentation, classification, and portion estimation.

# 1.9 Limitations and Important Notes

Although the NutriVision system provides a strong and practical approximation of nutritional content, the output should be interpreted as an estimate rather than an exact measurement.

## Known Limitations

The following limitations should be considered during use:

- Visually similar dishes may occasionally be confused by the classifier
- Similar variants of a food category may be difficult to distinguish in some cases
- Very small food items, side condiments, or garnishes may have lower portion-estimation accuracy
- Overlapping food items can reduce segmentation quality and affect final estimates
- If the ₹10 coin is missing, cropped, or unclear, the weight estimation becomes less reliable
- Nutritional values may vary depending on ingredients, oil quantity, cooking method, and recipe style
- Container shape (e.g., bowls or deep dishes) may influence the geometric assumptions used in volume estimation

## Practical Interpretation

Users should treat the output as a smart nutrition estimation tool that is useful for:

- approximate calorie tracking
- macro-nutrient awareness
- comparing meals
- making healthier dietary decisions

It is not intended to replace laboratory-grade nutritional analysis or professional dietary consultation.

# 4.10 Conclusion

The NutriVision Food Nutrition Estimator is designed to provide a simple, accessible, and transparent user experience for nutritional estimation from food images. By requiring only a clear food image and a visible ₹10 coin for calibration, the system can automatically:

- segment food items
- identify the dish
- estimate portion size and weight
- compute nutritional values
- present interpretable reasoning for the estimate

The deployed web interface is lightweight, easy to use, and well suited for both demonstration and practical evaluation. Under proper input conditions, the application offers an effective way to estimate the nutritional profile of Indian meals from a single photograph.